

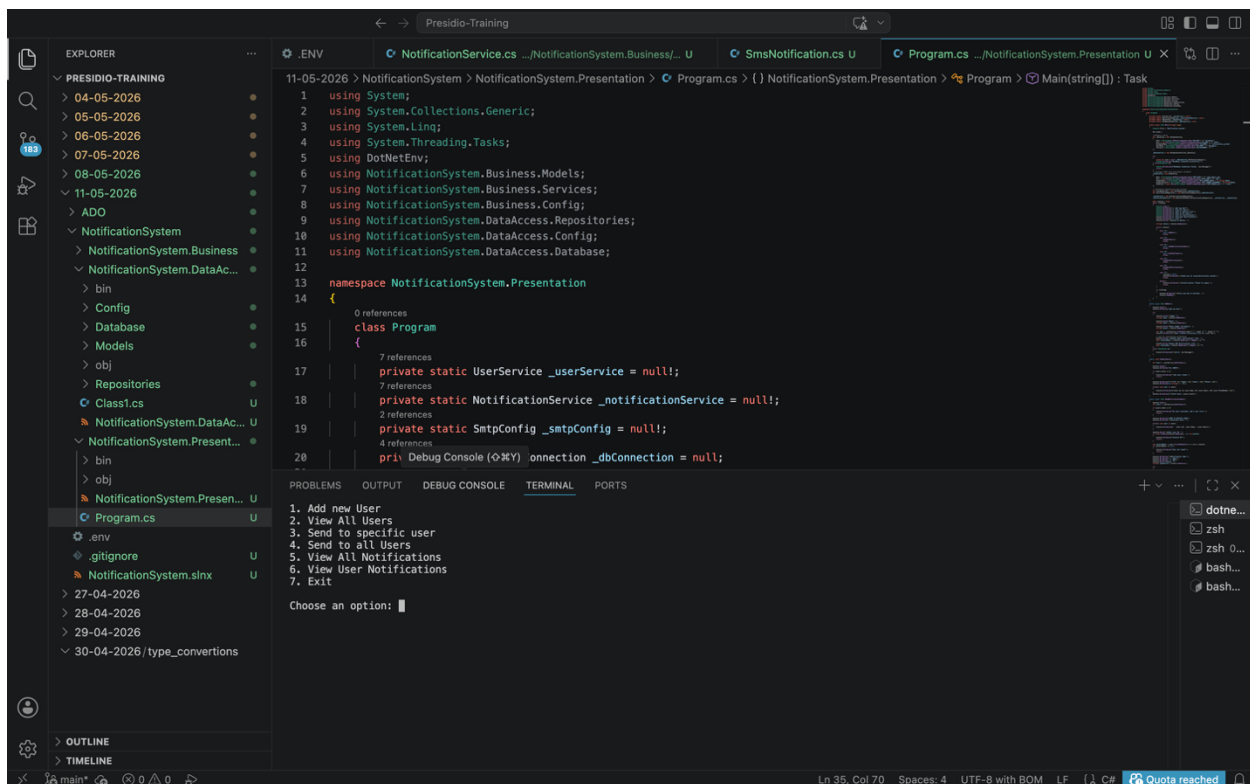
Date: 11-05-2026

GCR Assignment – Changing the Data Tier in Notification System to work with Database (postgres)

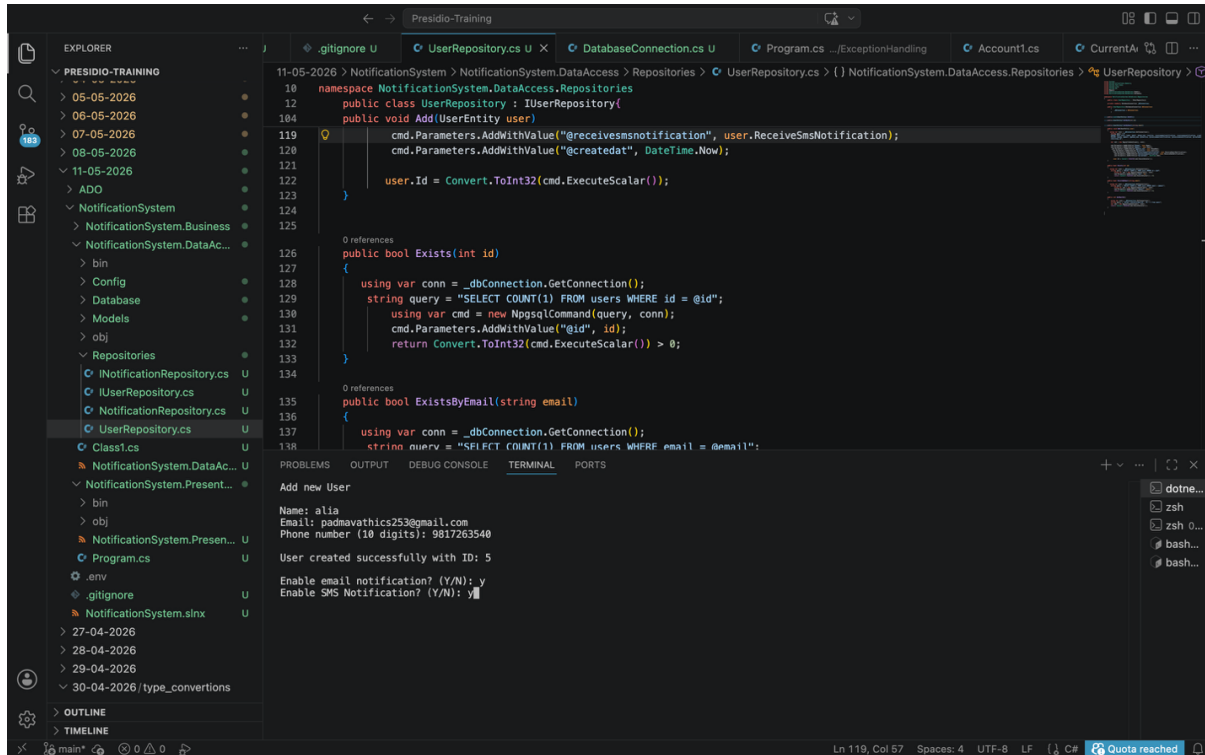
Source Code GitHub Url: <https://github.com/Padmavathi-SJ/Presidio-training/tree/main/11-05-2026/NotificationSystem>

Outputs:

Menu to choose operation:



Add new user:



```
11-05-2026 > NotificationSystem > NotificationSystem.DataAccess > Repositories > UserRepository.cs > {} NotificationSystem.DataAccess.Repositories > UserRepository >
10 namespace NotificationSystem.DataAccess.Repositories
11 public class UserRepository : IUserRepository{
12     public void Add(UserEntity user)
13     {
14         cmd.Parameters.AddWithValue("@receivesmsnotification", user.ReceiveSmsNotification);
15         cmd.Parameters.AddWithValue("@createdat", DateTime.Now);
16         user.Id = Convert.ToInt32(cmd.ExecuteScalar());
17     }
18 }
19 0 references
20 public bool Exists(int id)
21 {
22     using var conn = _dbConnection.GetConnection();
23     string query = "SELECT COUNT(1) FROM users WHERE id = @id";
24     using var cmd = new NpgsqlCommand(query, conn);
25     cmd.Parameters.AddWithValue("@id", id);
26     return Convert.ToInt32(cmd.ExecuteScalar()) > 0;
27 }
28 0 references
29 public bool ExistsByEmail(string email)
30 {
31     using var conn = _dbConnection.GetConnection();
32     string query = "SELECT COUNT(1) FROM users WHERE email = @email";
33     cmd.Parameters.AddWithValue("@email", email);
34     return Convert.ToInt32(cmd.ExecuteScalar()) > 0;
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200 }
```

Problems OUTPUT DEBUG CONSOLE TERMINAL PORTS

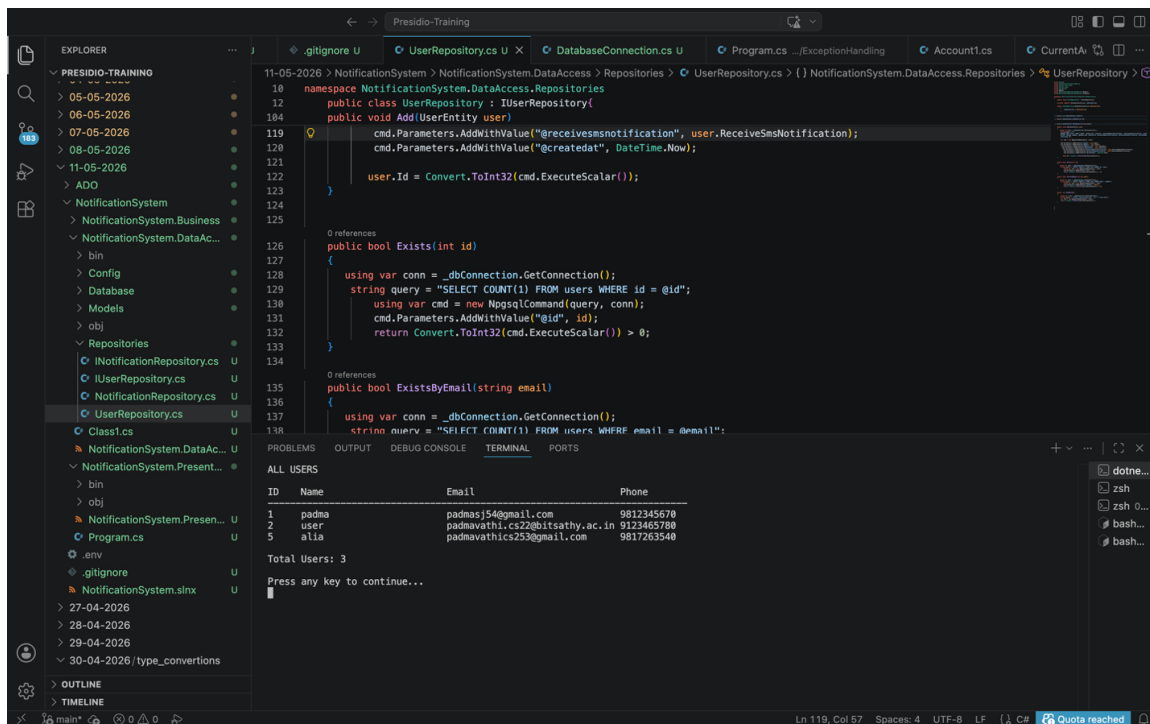
Add new User

Name: alia
Email: padnavathics253@gmail.com
Phone number (10 digits): 9817263540

User created successfully with ID: 5

Enable email notification? (Y/N): y
Enable SMS Notification? (Y/N): y

Display all active users:



```
11-05-2026 > NotificationSystem > NotificationSystem.DataAccess > Repositories > UserRepository.cs > {} NotificationSystem.DataAccess.Repositories > UserRepository >
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182 }
183 }
184 }
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190 }
```

Problems OUTPUT DEBUG CONSOLE TERMINAL PORTS

ALL USERS

ID	Name	Email	Phone
1	padma	padmas154@gmail.com	9812345678
2	user	padnavathi.cs22@itsathy.ac.in	9123465780
5	alia	padnavathics253@gmail.com	9817263540

Total Users: 3

Press any key to continue...

Users table with newly added user(id = 5) in notification_system database:

The screenshot shows the DBeaver 26.0.3 interface. The left sidebar displays the database structure for 'notification_system', including tables 'notifications' (96K) and 'users' (40K). The main window shows the 'users' table with the following data:

id	name	email	phone_num	isactive	receiveemailnotification	receivesmsnotification
1	padma	padmasj54@gmail.com	9812345670	[v]	[v]	[v]
2	user	padmavathi.cs22@bitsathy.ac.in	9123465780	[v]	[v]	[v]
5	alia	padmavathics253@gmail.com	9817263540	[v]	[v]	[v]

The SQL editor shows the following queries:

```
SELECT column_name, data_type, is_nullable, column_default FROM information_schema.columns WHERE table_name = 'notifications';
```

```
select * from users;
```

Send email notification to a specific user and save in notifications table:

The screenshot shows the Visual Studio Code interface with the 'UserRepository.cs' file open. The code defines a 'UserRepository' class with the following methods:

```
namespace NotificationSystem.DataAccess.Repositories
{
    public class UserRepository : IUserRepository
    {
        public void Add(UserEntity user)
        {
            cmd.Parameters.AddWithValue("@receivesmsnotification", user.ReceiveSmsNotification);
            cmd.Parameters.AddWithValue("@createdat", DateTime.Now);

            user.Id = Convert.ToInt32(cmd.ExecuteScalar());
        }

        public bool Exists(int id)
        {
            using var conn = _dbConnection.GetConnection();
        }
    }
}
```

The terminal output shows the execution of the 'Add' method, which successfully adds a user and sends an email notification:

```
SEND TO SPECIFIC USER
Available Users:
1. padma - padmasj54@gmail.com
2. user - padmavathi.cs22@bitsathy.ac.in
5. alia - padmavathics253@gmail.com
Enter user ID: 5
Notification Type:
1. Email
2. SMS
Choose: 1
Subject: this subject is for user with user_id 5
Message: this message is for user with user_id 5
Email sent successfully:
To: padmavathics253@gmail.com
Subject: this subject is for user with user_id 5
Time: 23:55:38
Notification sent successfully to alia!
Press any key to continue...
```

Sent notifications data in notifications table:

The screenshot shows the DBeaver 26.0.3 interface. The left sidebar displays the database structure for 'public' schema, including tables 'notifications' (96K) and 'users' (40K). The SQL editor contains the following query:

```
SELECT
  column_name,
  data_type,
  is_nullable,
  column_default
FROM information_schema.columns
WHERE table_name = 'notifications';

select *from users;

select *from notifications;
```

The 'notifications' table data is displayed in a grid view:

	id	user_id	user_name	type	subject	message	recipient	is_sent
1	1	1	padma	Email	this is test-2	this message is test-2	padmasj54@gmail.com	[v]
2	2	2	user	Email	this is test-2	this message is test-2	padmavathi.cs22@bitsathy.ac.in	[v]
3	3	5	alia	Email	this subject is for user with	this message is for user with u	padmavathics253@gmail.com	[v]

Email notification sent to user(id=5)

The screenshot shows an email notification on a mobile device. The email is from 'padmavathics2005@gmail.com' to 'padmavathics253@gmail.com', dated 'May 11, 2026, 11:55 PM'. The subject is 'this subject is for user with user_id 5'. The email body contains the following text:

Dear alia,
this message is for user with user_id 5

Sent at: 2026-05-11 23:55:34
Notification System

Get and view all sent notifications:

The screenshot shows the Visual Studio Code editor with the `UserRepository.cs` file open. The file contains the following code:

```
11-05-2026 > NotificationSystem > NotificationSystem.DataAccess > Repositories > UserRepository.cs > {} NotificationSystem.DataAccess.Repositories > UserRepository >
10 namespace NotificationSystem.DataAccess.Repositories
12 public class UserRepository : IUserRepository{
184 public void Add(UserEntity user)
119 {
120     cmd.Parameters.AddWithValue("@receivesmsnotification", user.ReceiveSmsNotification);
120     cmd.Parameters.AddWithValue("@createdat", DateTime.Now);
121     user.Id = Convert.ToInt32(cmd.ExecuteScalar());
122 }
123
124 0 references
126 public bool Exists(int id)
127 {
128     using var conn = _dbConnection.GetConnection();
```

The terminal output shows the results of the `GetAllSentNotifications` method:

```
ALL NOTIFICATIONS
ID Type To Message Status
-----
1 Email padmasj54@gmail.com this is test-2 Sent at 23:33
2 Email padmavathi.cs22@bitsathy.ac.in this is test-2 Sent at 23:33
3 Email padmavathics253@gmail.com this subject is for user with user_id 5 Sent at 23:55
Press any key to continue...
```

Get and view all sent notifications of a user:

The screenshot shows the Visual Studio Code editor with the `UserRepository.cs` file open. The file contains the following code:

```
11-05-2026 > NotificationSystem > NotificationSystem.DataAccess > Repositories > UserRepository.cs > {} NotificationSystem.DataAccess.Repositories > UserRepository >
10 namespace NotificationSystem.DataAccess.Repositories
12 public class UserRepository : IUserRepository{
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119 {
120     cmd.Parameters.AddWithValue("@receivesmsnotification", user.ReceiveSmsNotification);
120     cmd.Parameters.AddWithValue("@createdat", DateTime.Now);
121     user.Id = Convert.ToInt32(cmd.ExecuteScalar());
122 }
123
124 0 references
126 public bool Exists(int id)
127 {
128     using var conn = _dbConnection.GetConnection();
```

The terminal output shows the results of the `GetUserSentNotifications` method:

```
USER NOTIFICATIONS
Select User:
1. padma
2. user
5. alia
Enter user ID: 5
Notifications for user ID 5:
[3] Email: Subject: this subject is for user with user_id 5
Sent at 2026-05-11 23:55
Total: 1 notifications
Press any key to continue...
```